



VISUAL ANALYTICS: NETWORK ANALYSIS

AB MOSCA (THEY/THEM)



ANNOUNCEMENTS / REMINDERS

- Thursday 02/26 is Rally Day
 - No class
 - Make up office hours 10:30-12:00
 - Mid-semester project released – start early
- Next week (03/03 & 03/05)
 - Geospatial analysis with SAL
 - 03/03 in our regular classroom
 - 03/05 in the SAL (Sabin-Reed 104)

TODAY'S PLAN

- Warm up
- Data
- Visualization

WARM UP: GRAPH DATA

1. What is the mathematical definition of graph data?
2. What does graph data model?
3. What insights might we gain from analyzing graph data?
4. Name two ways to visualize graph data. What are the pros and cons to each?

DATA

- We will work with the Les Misérables Co-Occurrence dataset today
 - Raw Data: <https://gist.github.com/abkunal/98d35b9b235312e90f3e43c9f7b6932b>
 - Example Vis: <https://bost.ocks.org/mike/miserables/>
 - Wiki: https://en.wikipedia.org/wiki/Les_Mis%C3%A9rables
- Familiarize yourself with the data:
 1. How is the raw data formatted?
 2. What information does the raw data contain?
 3. What questions do you have about the data?

D3: REQUESTING DATA

- Typically, the data we visualize will be in an external file or database
- To load external data, your program must make a **request** (<https://www.d3indepth.com/requests/>):

Here's an example of a JSON file that's located at <https://assets.codepen.io/2814973/my-json.json>:

```
[
  {
    "name": "Paris",
    "indicator1": 4030,
    "indicator2": 13.45
  },
  {
    "name": "Tokyo",
    "indicator1": 3912,
    "indicator2": 45.41
  },
  {
    "name": "New York",
    "indicator1": 19481,
    "indicator2": 32.53
  }
]
```

You request a JSON file using `json` from the `d3-fetch` module:

```
json('https://assets.codepen.io/2814973/my-json.json')
  .then(function(data) {
    console.log(data);
  });
```

In this example, the JSON file represents an array, so you can join it to HTML or SVG elements:

```
function update(data) {
  select('#content tbody')
    .selectAll('tr')
    .data(data)
    .join('tr')
    .html(function(d) {
      let html = '<tr>';
      html += '<td>' + d.name + '</td>';
      html += '<td>' + d.indicator1 + '</td>';
      html += '<td>' + d.indicator2 + '</td>';
      html += '</tr>';
      return html;
    });
}

json('https://assets.codepen.io/2814973/my-json.json')
  .then(function(data) {
    update(data);
  });
```

Name	Indicator 1	Indicator 2
Paris	4030	13.45
Tokyo	3912	45.41
New York	19481	32.53

YOUR TURN

- Create a folder and add starter html, css, and js files to it. Add the D3 library to your folder
- Download the Les Misérables Co-Occurrence data and save it in the same folder as your starter files
- Use d3 to **request the Les Mis data** and **print the data** to the console

YOUR TURN

- Review the D3 overview for force-directed layouts: <https://www.d3indepth.com/force-layout/>
- Write a program to visualize the Les Mis data with a force directed layout

